Mean, Median and the Five-Point Summary

Name _____ Date _____ Calculator allowed Version 1

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KEY WORDS mean *(toharite)* · median *(tau waenga)* · lower quartile *(hauwhā raro)*

A The mean ●○○

Remember: Mean = the total of all the values ÷ the number of values.

1. Find the mean of each set of data.

a) 4, 7, 9, 12

b) 15, 20, 22, 18, 25

c) 3, 3, 5, 8, 9, 2

d) 2.5, 4, 3.5, 6

e) 12, 0, 7, 9, 15, 11, 9

f) 48, 52, 55, 61

g) 1.2, 1.8, 2.4, 0.6, 1.5

h) 100, 85, 90, 110, 95, 120

B The median ●○○

Remember: Put the values in order first. With an even number of values, the median is halfway between the middle two.

2. Find the median of each set of data.

a) 7, 3, 9, 5, 12

b) 4, 8, 6, 10

c) 15, 9, 22, 13, 18, 11, 20

d) 2, 9, 4, 7, 7, 3

e) 31, 25, 40, 28, 36, 33, 25, 29

f) 1.5, 0.8, 2.1, 1.2, 1.9

C Quartiles ●●○

Remember: The lower quartile (LQ) is the median of the lower half of the data. The upper quartile (UQ) is the median of the upper half. If there is a middle value, leave it out of both halves.

3. These sets of data are already in order. Find the lower quartile and the upper quartile.

a) 2, 4, 5, 7, 8, 10, 13

b) 1, 3, 4, 6, 9, 11, 12, 15

c) 10, 12, 15, 16, 18, 20, 23, 25, 30

d) 5, 6, 8, 8, 9, 12, 14, 15, 17, 20

e) 21, 24, 26, 30, 33, 35, 36, 40, 41, 45, 48

f) 3, 7, 8, 11, 14, 18

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D The five-point summary ●●●

> **Remember:** Minimum, lower quartile, median, upper quartile, maximum. Range = maximum − minimum. Interquartile range = UQ − LQ.

4. The numbers of text messages eleven students sent in one day: 12, 30, 8, 25, 17, 40, 22, 15, 9, 33, 20

a) Find the five-point summary.

b) Find the range.

c) Find the interquartile range.

5. The times, in minutes, that ten students took to get to school: 15, 22, 8, 35, 12, 18, 40, 25, 10, 20

a) Find the five-point summary.

b) Find the range.

c) Find the interquartile range.

6. The heights, in centimetres, of nine tomato plants: 34, 41, 28, 45, 39, 52, 36, 30, 47

a) Find the five-point summary.

b) Find the range.

c) Find the interquartile range.

E Mean or median? ●●●

> ⚠ **Watch out:** One value that is far from the rest changes the mean a lot, but it hardly changes the median.

7. The weekly pay, in dollars, of seven workers at a small company: 800, 850, 820, 900, 870, 840, 3,200

a) Find the mean.

b) Find the median.

c) Which describes a typical worker's pay better, the mean or the median? Give a reason.

8. The marks, out of 50, of eight students in a test: 38, 42, 35, 40, 44, 4, 39, 41

a) Find the mean.

b) Find the median.

c) Which describes how the class did better, the mean or the median? Give a reason.

9. The ages, in years, of the seven people at a birthday party: 6, 7, 6, 8, 7, 6, 37

a) Find the mean.

b) Find the median.

c) Which describes the age of a typical guest better, the mean or the median? Give a reason.

★ Extension

10. The mean of five numbers is 12. Four of the numbers are 8, 15, 10, 13. Find the fifth number.

11. The mean of 6 test scores is 70. One more score of 84 is added. Find the new mean.

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Mean, Median and the Five-Point Summary

Name _____ Date _____ Calculator allowed Version 2

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KEY WORDS mean *(toharite)* · median *(tau waenga)* · lower quartile *(hauwhā raro)*

A The mean ●○○

Remember: Mean = the total of all the values ÷ the number of values.

1. Find the mean of each set of data.

a) 5, 8, 6, 13

b) 14, 21, 19, 16, 30

c) 4, 4, 6, 7, 9, 6

d) 1.5, 5, 4.5, 7

e) 10, 0, 8, 6, 13, 12, 7

f) 42, 56, 53, 65

g) 2.2, 1.6, 2.8, 0.4, 2.5

h) 105, 80, 95, 115, 90, 85

B The median ●○○

Remember: Put the values in order first. With an even number of values, the median is halfway between the middle two.

2. Find the median of each set of data.

a) 8, 2, 10, 6, 13

b) 5, 9, 7, 12

c) 14, 8, 21, 17, 19, 10, 23

d) 3, 8, 5, 6, 6, 2

e) 32, 26, 41, 27, 35, 34, 26, 30

f) 1.4, 0.9, 2.3, 1.1, 1.8

C Quartiles ●●○

Remember: The lower quartile (LQ) is the median of the lower half of the data. The upper quartile (UQ) is the median of the upper half. If there is a middle value, leave it out of both halves.

3. These sets of data are already in order. Find the lower quartile and the upper quartile.

a) 1, 3, 6, 8, 9, 12, 14

b) 2, 4, 5, 7, 10, 12, 13, 17

c) 11, 13, 14, 17, 19, 22, 24, 27, 31

d) 4, 7, 7, 9, 10, 13, 15, 16, 18, 22

e) 20, 23, 27, 31, 32, 34, 38, 39, 42, 46, 49

f) 2, 6, 9, 12, 15, 19

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D The five-point summary ●●●

> **Remember:** Minimum, lower quartile, median, upper quartile, maximum. Range = maximum − minimum. Interquartile range = UQ − LQ.

4. The numbers of text messages eleven students sent in one day: 14, 32, 6, 27, 19, 42, 24, 11, 9, 35, 21

a) Find the five-point summary.

b) Find the range.

c) Find the interquartile range.

5. The times, in minutes, that ten students took to get to school: 14, 24, 6, 32, 10, 20, 45, 28, 12, 18

a) Find the five-point summary.

b) Find the range.

c) Find the interquartile range.

6. The heights, in centimetres, of nine tomato plants: 32, 43, 26, 48, 37, 54, 35, 29, 44

a) Find the five-point summary.

b) Find the range.

c) Find the interquartile range.

E Mean or median? ●●●

> ⚠ **Watch out:** One value that is far from the rest changes the mean a lot, but it hardly changes the median.

7. The weekly pay, in dollars, of seven workers at a small company: 750, 820, 790, 860, 800, 780, 2,900

a) Find the mean.

b) Find the median.

c) Which describes a typical worker's pay better, the mean or the median? Give a reason.

8. The marks, out of 50, of eight students in a test: 36, 43, 37, 41, 45, 6, 40, 38

a) Find the mean.

b) Find the median.

c) Which describes how the class did better, the mean or the median? Give a reason.

9. The ages, in years, of the seven people at a birthday party: 5, 6, 7, 6, 5, 7, 41

a) Find the mean.

b) Find the median.

c) Which describes the age of a typical guest better, the mean or the median? Give a reason.

★ Extension

10. The mean of five numbers is 14. Four of the numbers are 9, 17, 12, 16. Find the fifth number.

11. The mean of 5 test scores is 60. One more score of 78 is added. Find the new mean.

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